

SAMARTHA H M

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Electronics and Communication Engineer — Hardware Design, IoT Automation, Prototyping & Testing

Electronics and Communication Engineering graduate with hands-on experience in PCB design, real-time firmware development, and hardware-software integration. Proven ability to build and deploy complete embedded and robotic systems and take products from concept to market within a startup environment. Strong interest in embedded systems, autonomous robotics, and IoT automation.

Education

Srinivas Institute of Technology

B.E. Electronics & Communication Engineering – CGPA: 7.86/10

Expected Jun 2026

Valachil, Mangalore

Experience

Experimind Labs Pvt. Ltd.

Feb 2026 – May 2026

Core Team Member (Promoted from Electronics Intern within 2 months)

- Contributed to R&D and full product cycle — ideation, PCB design, prototyping, SMD/through-hole assembly, testing, and market release — for two educational IoT kits delivered to 300+ students across 100+ production units.
- Designed Kit 1 (14-component sensor board: 11 input sensors, 3 output actuators, custom PCB, JST wiring, battery pack) and Kit 2 (2-motor line follower robot platform with IR/LDR sensing) entirely from scratch with 98%+ first-pass assembly quality.
- Developed embedded control system for Raspberry Pi Pico W-based autonomous land survey robot with waypoint and free-path navigation using GPS, magnetometer, and gyroscope — prototype built and demonstrated.

ISPARK Learning Solutions

Feb – May 2026

Embedded Systems & IoT Intern (Remote)

- Built AI-powered smart waste segregation prototype in 5 days: ESP32 (sensors + servo) + ESP32-CAM + Python MobileNetV2 classifier (79% accuracy on 600-image dataset) with custom serial protocol achieving sub-second DRY/WET/SKIP sorting response.
- Designed complete hardware, firmware, and Python inference engine with confidence thresholding and real-time telemetry dashboard.

Projects

Autonomous Arecanut Drying Robot | Final Year Capstone Project (Team Lead)

- Designed 5-motor 4WD platform ($\pm 5\text{cm}$ accuracy on $5\times 5\text{m}$ grid) with dual-controller architecture (Raspberry Pi 5 + Arduino Mega) and custom binary protocol achieving $<50\text{ms}$ command latency for reliable real-time teleoperation.
- Designed BTS7960 motor drive circuitry (40A peak load) and implemented grid-based serpentine traversal algorithm ensuring complete autonomous area coverage across agricultural drying yards.
- Addressed post-harvest loss in arecanut farming by replacing labor-intensive manual raking with a fully autonomous platform; built self-hosted Wi-Fi access point enabling smartphone control without internet dependency — critical for rural deployment.
- Architected end-to-end system integration across hardware, real-time firmware, ML pipeline, and React frontend — ensuring all subsystems operated reliably as a single cohesive system under real field conditions.
- Built React dashboard with dark mode, hold-to-move safety mechanism, and real-time telemetry visualization. Led 4-member cross-functional team from design through field deployment and demonstration.

One-Way Vehicle Direction Monitor | Embedded Safety System

- Built dual-lane wrong-way detection model (95% accuracy) using HC-SR04 sensors with instant servo barrier activation and audio-visual alerts; programmed Arduino Uno with concurrent dual-lane sensor logic.

Personal Projects & Automation Tools | Independent Builds

- Shipped SyncPadIO (real-time text sharing), WebsiteON (serverless monitor + Telegram alerts), SonicSync (WebRTC audio), YotuDrive (Reed-Solomon encoding), Mobtola Pro (ADB backup) — Python, React, Node.js, TypeScript.

Technical Skills

Programming: Python, C/C++, JavaScript/TypeScript, Arduino IDE

Embedded Systems: Arduino (UNO/Mega), Raspberry Pi (5/Pico W), ESP32, Real-Time Control, PWM, Motor Control, Sensor Integration

Circuit Design & PCB: KiCad, EasyEDA, LTspice, BTS7960 Drivers, Power Electronics, SMD/Through-Hole Soldering, PCB Layout

AI & IoT: MobileNetV2, OpenCV, Computer Vision, IoT System Integration, Real-Time Data Acquisition

Tools & Workflow: Git, VS Code, Linux, Adobe Premiere Pro, Canva, Microsoft Office

Leadership & Extracurriculars

Event Organizer | ENVISION 2k25 National Tech Fest

- Organized Line Follower robotics competition for 100+ participants and led promotional design team.

Editor & Designer | STEADY Department Club (2024 & 2025)

- Designed 50-page technical magazine distributed to 500+ students and faculty. Organized 50+ technical and non-technical college events.

Active Member & Junior Mentor | TECHBOTS Robotics Club

- Mentored junior students in embedded programming and PCB design. Contributed to robotics systems development.

Content Creator | DreamBeatz YouTube Channel

- Music and lyrical edit content with 3K+ subscribers.